

EarthCube Oceanography and
Geobiology Environmental 'Omics



ECOGEO Workshop 2: Introduction to Env 'Omics

Amplicon Analysis - QIIME

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THE UNIVERSITY OF
CHICAGO

Introduction – Quantitative Insights Into Microbial Ecology (QIIME)

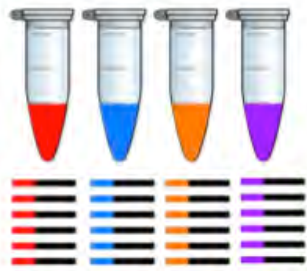
- Open source software package for taxonomic analysis of 16S rRNA sequences
- UC Colorado & Northern Arizona University
- www.qiime.org (great resource.....)
- Good community support
- Can  most problems
- Multi-platform



Greg Caporaso



Rob Knight



Extract DNA and
amplify marker gene
with barcoded
primers

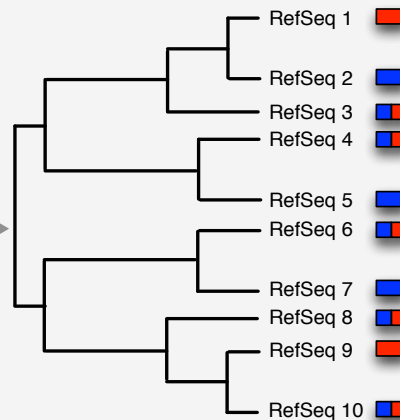


Pool amplicons and sequence

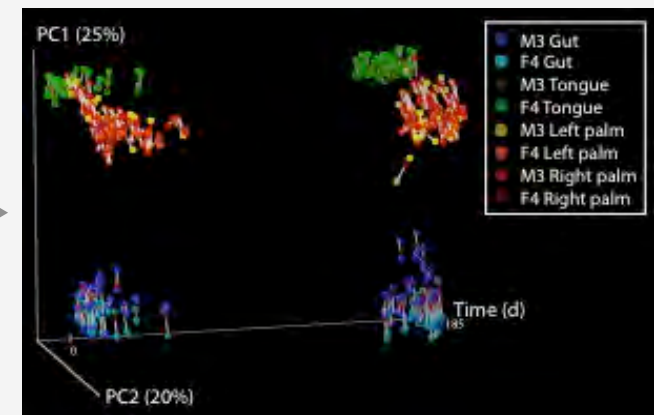


>GCACCTGAGGACAGGCATGAGGAA...
>GCACCTGAGGACAGGGGAGGAGGA...
>TCACATGAACCTAGGCAGGACGAA...
>CTACCGGAGGACAGGCATGAGGAT...
>TCACATGAACCTAGGCAGGAGGAA...
>GCACCTGAGGACACGCAGGACGAC...
>CTACCGGAGGACAGGCAGGAGGAA...
>CTACCGGAGGACACACAGGAGGAA...
>GAACCTTCACATAGGCAGGAGGAT...
>TCACATGAACCTAGGGGCAAGGAA...
>GCACCTGAGGACAGGCAGGAGGAA...

Assign reads to
samples



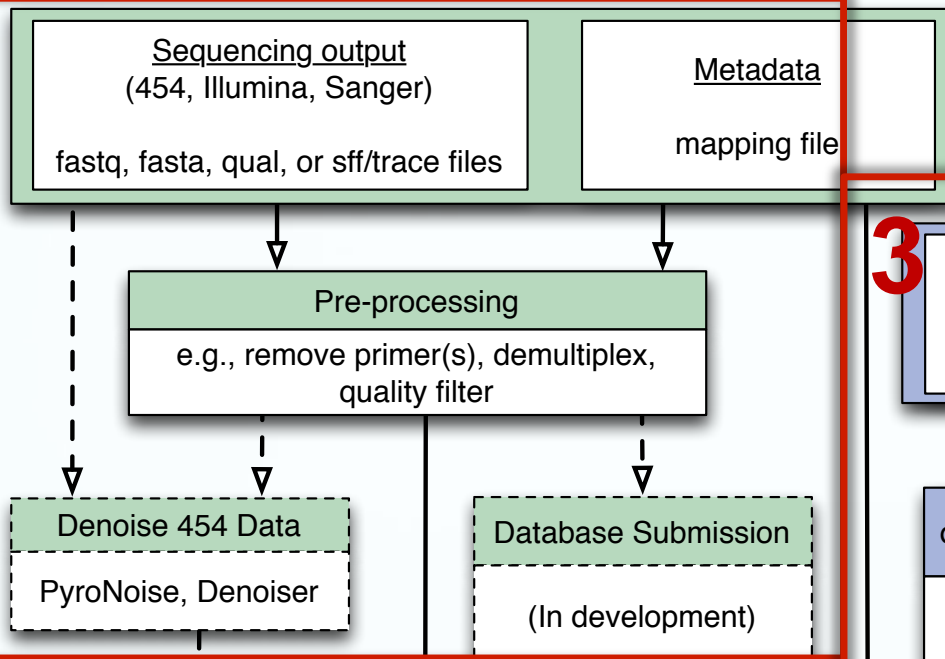
Assign millions of
sequences from
thousands of samples
to OTUs



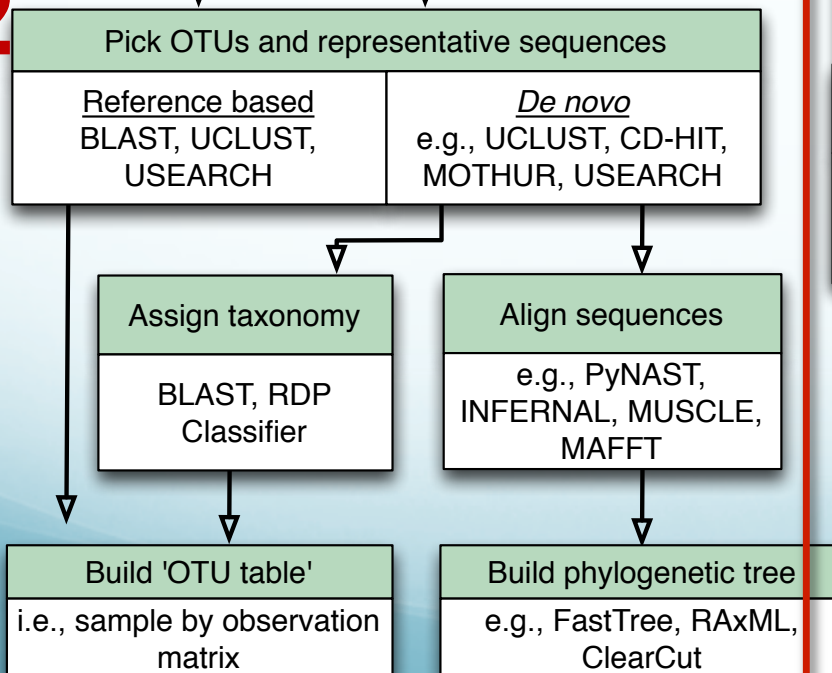
Compare samples

www.QIIME.org

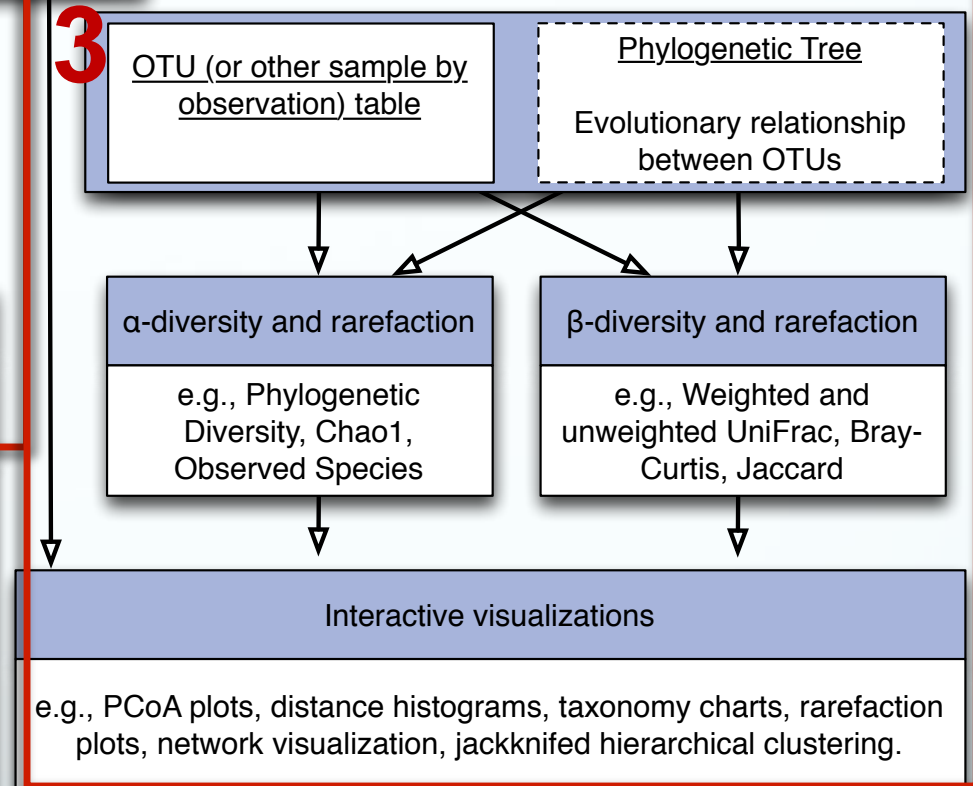
1



2



3



Legend

Currently supported for
marker-gene data only

(i.e., 'upstream' step)

Required step or input

Currently supported for
general sample by
observation data

(i.e., 'downstream' step)

Optional step or input

Before we start – Key QIIME files

- Mapping file:
 - Tab-delimited text file (.txt)
 - Per sample meta-data, user-defined
- Input sequence file:
 - Raw files from sequencing machine
e.g. PE reads from Illumina Miseq
Undetermined_S0_L001_R1_001.fastq
Undetermined_S0_L001_R2_001.fastq
Undetermined_S0_L001_I1_001.fastq
- OTU table (for all downstream analysis):
 - sample x OTU matrix (.txt file)
 - BIOM file (.biom)

Contents of ~/ecogeo/qiimedir

1. gg_13_8_otu: Greengenes database
2. Practice:
 - map_file.txt → Mapping file
 - subs_seqs_q.fastq, subs_seqs_q.fasta → Demultiplexed, quality-filtered sequence files
 - Three biom files → Three OTU picking strategies
 - scripts.txt → Steps performed by me
 - alpha_params.txt → Parameters for alpha diversity
3. Precomputed: Output of OTU picking steps + Results

1. Checking mapping file format

```
$ validate_mapping_file.py -m map_file.txt -o mapping_file_output
```

Name (ID) of sample

Primer

Sample categories (treatments)

#SampleID	BarcodeSequence	LinkerPrimerSequence	Treatment	DOB	Description
PC.354	AGCACGAGCCTA	YATGCTGCCTCCCGTAGGAGT	Control	20061218	Control_mouse_I.D._354
PC.355	AACTCGTCGATG	YATGCTGCCTCCCGTAGGAGT	Control	20061218	Control_mouse_I.D._355
PC.356	ACAGACCACTCA	YATGCTGCCTCCCGTAGGAGT	Control	20061126	Control_mouse_I.D._356
PC.481	ACCAGCGACTAG	YATGCTGCCTCCCGTAGGAGT	Control	20070314	Control_mouse_I.D._481
PC.593	AGCAGCACTTGT	YATGCTGCCTCCCGTAGGAGT	Control	20071210	Control_mouse_I.D._593
PC.607	AACTGTGCGTAC	YATGCTGCCTCCCGTAGGAGT	Fast	20071112	Fasting_mouse_I.D._607
PC.634	ACAGAGTCGGCT	YATGCTGCCTCCCGTAGGAGT	Fast	20080116	Fasting_mouse_I.D._634
PC.635	ACCGCAGAGTCA	YATGCTGCCTCCCGTAGGAGT	Fast	20080116	Fasting_mouse_I.D._635
PC.636	ACGGTGAGTGTC	YATGCTGCCTCCCGTAGGAGT	Fast	20080116	Fasting_mouse_I.D._636

Line: 8 Column: 33 Rest Tab Size: 4

2.1 Join paired end reads

- Joins forward and reverse reads
- QIIME default: fastq_join (min. overlap = 6 bp and min. % identity = 8%)

```
$ join_paired_ends.py -f Undetermined_S0_L001_R1_001.fastq -r  
Undetermined_S0_L001_R2_001.fastq -b  
Undetermined_S0_L001_I1_001.fastq -o reads
```

Helpful hint: FastQC forward & reverse reads before joining

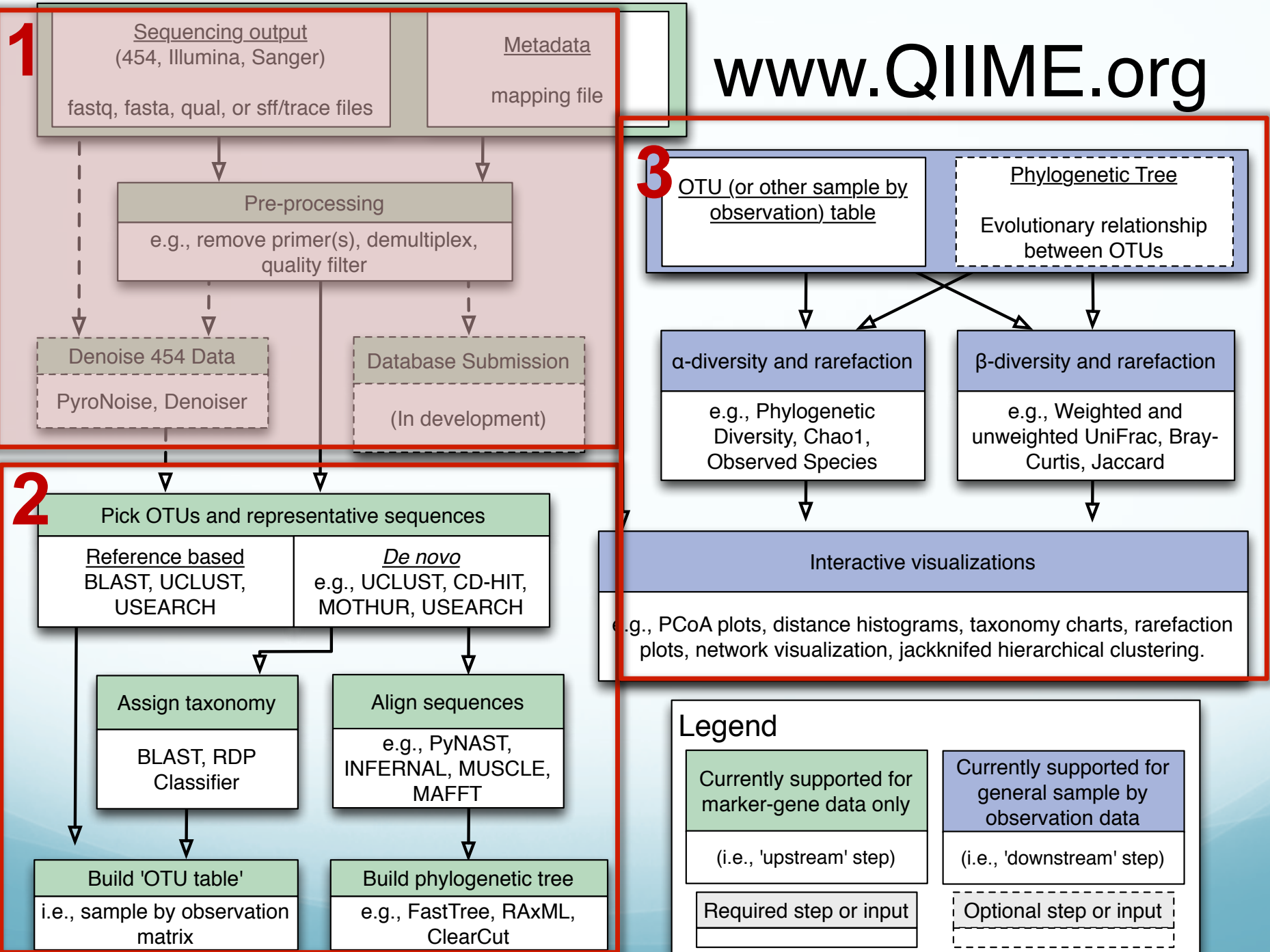
2.2 Demultiplex & quality filter

- Using sample specific barcodes, identify each sequence with a sample
- Performs QC:
 - Removes sequences <75% consecutive high quality base calls
 - Removes sequences with a quality score <29
 - Removes sequences with N
 - Truncate sequences with >3 consecutive low quality base calls
 - Produces seqs.fna, seqs.fastq (optional), histogram.txt, split_library_log.txt

```
$ split_libraries_fastq.py -i fastqjoin.join.fastq -b  
fastqjoin.join_barcodes.fastq -o split -m map_file.txt -q 29  
--barcode_type 12 --store_demultiplexed_fastq
```

Helpful hint: Be mindful of preprocessing reads prior to demultiplexing in Qiime

www.QIIME.org

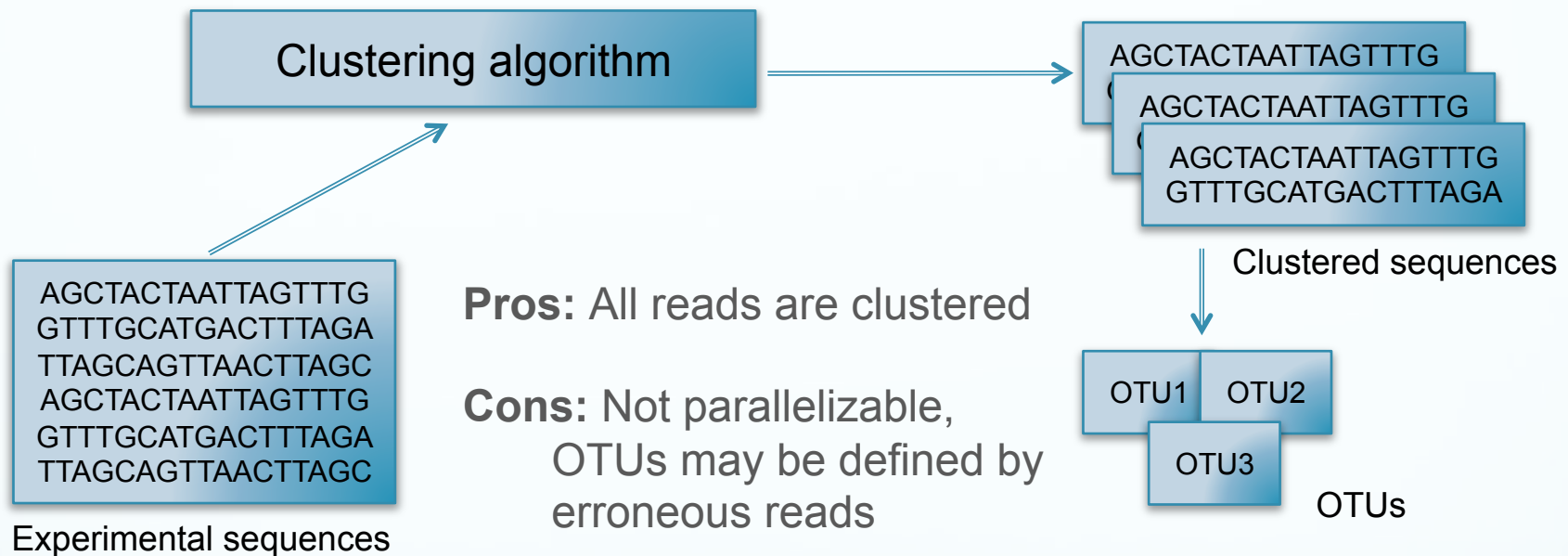


3. Picking OTUs (with workflow scripts)

- *De novo* (clustering of sequences at 97%)
- Closed-reference
- Open-reference

Helpful hint: Know the defaults.

3.1 *De novo* OTU picking



```
$ pick_de_novo_otus.py -i subs_seqs_q.fasta -o uclust_otus/
```

Workflow script: pick OTUs, align sequences, assign taxonomy, construct phylogenetic tree, and construct OTU table

3.1 *De novo* OTU picking contd.

```
$ pick_de_novo_otus.py -i subs_seqs_q.fasta -o uclust_otus/
```

`pick_otus.py` – determine OTU clusters

`pick_rep_set.py` – pick representative sequence for each OTU cluster

`align_seqs.py` – align sequences to a template or a reference alignment

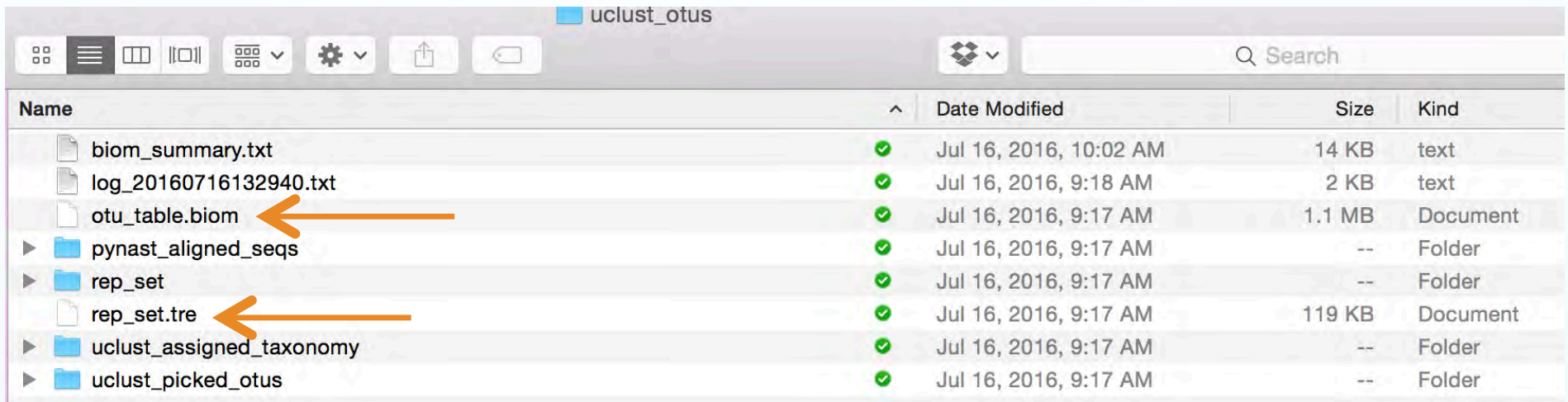
`assign_taxonomy.py` – assign taxonomy to representative sequences

`filter_alignment.py` – remove non-phylogenetically informative positions

`make_phylogeny.py` – construct phylogenetic tree

`make_otu_table.py` – make actual OTU table object (.biom)

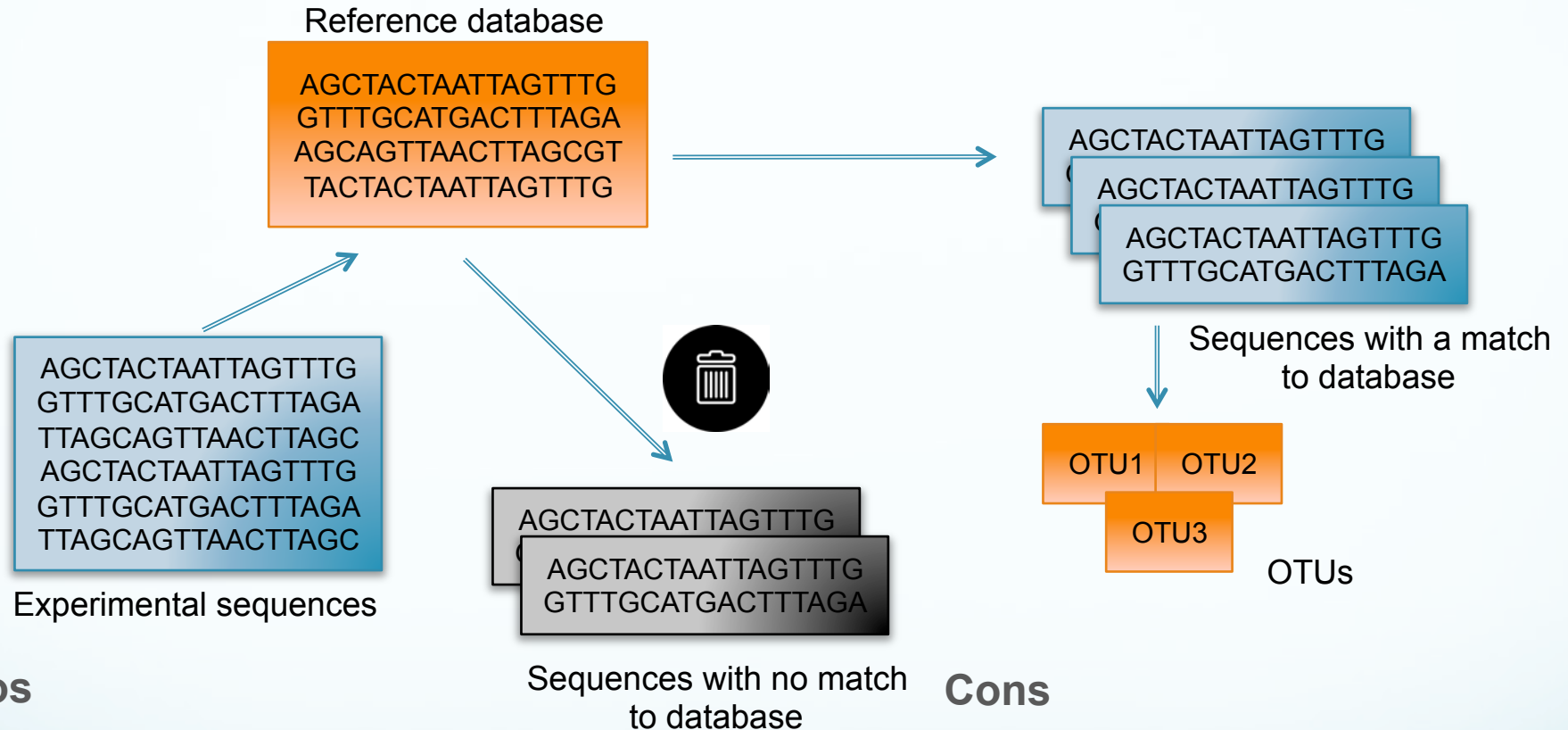
3.1 *De novo* OTU picking



The screenshot shows a file explorer window titled "uclust_otus". The window displays a list of files and folders with columns for Name, Date Modified, Size, and Kind. Two orange arrows point to the files "otu_table.biom" and "rep_set.tre".

Name	✓	Date Modified	Size	Kind
biom_summary.txt	✓	Jul 16, 2016, 10:02 AM	14 KB	text
log_20160716132940.txt	✓	Jul 16, 2016, 9:18 AM	2 KB	text
otu_table.biom	✓	Jul 16, 2016, 9:17 AM	1.1 MB	Document
▶ pynast_aligned_seqs	✓	Jul 16, 2016, 9:17 AM	--	Folder
▶ rep_set	✓	Jul 16, 2016, 9:17 AM	--	Folder
rep_set.tre	✓	Jul 16, 2016, 9:17 AM	119 KB	Document
▶ uclust_assigned_taxonomy	✓	Jul 16, 2016, 9:17 AM	--	Folder
▶ uclust_picked_otus	✓	Jul 16, 2016, 9:17 AM	--	Folder

3.2 Closed-reference OTU picking



Pros

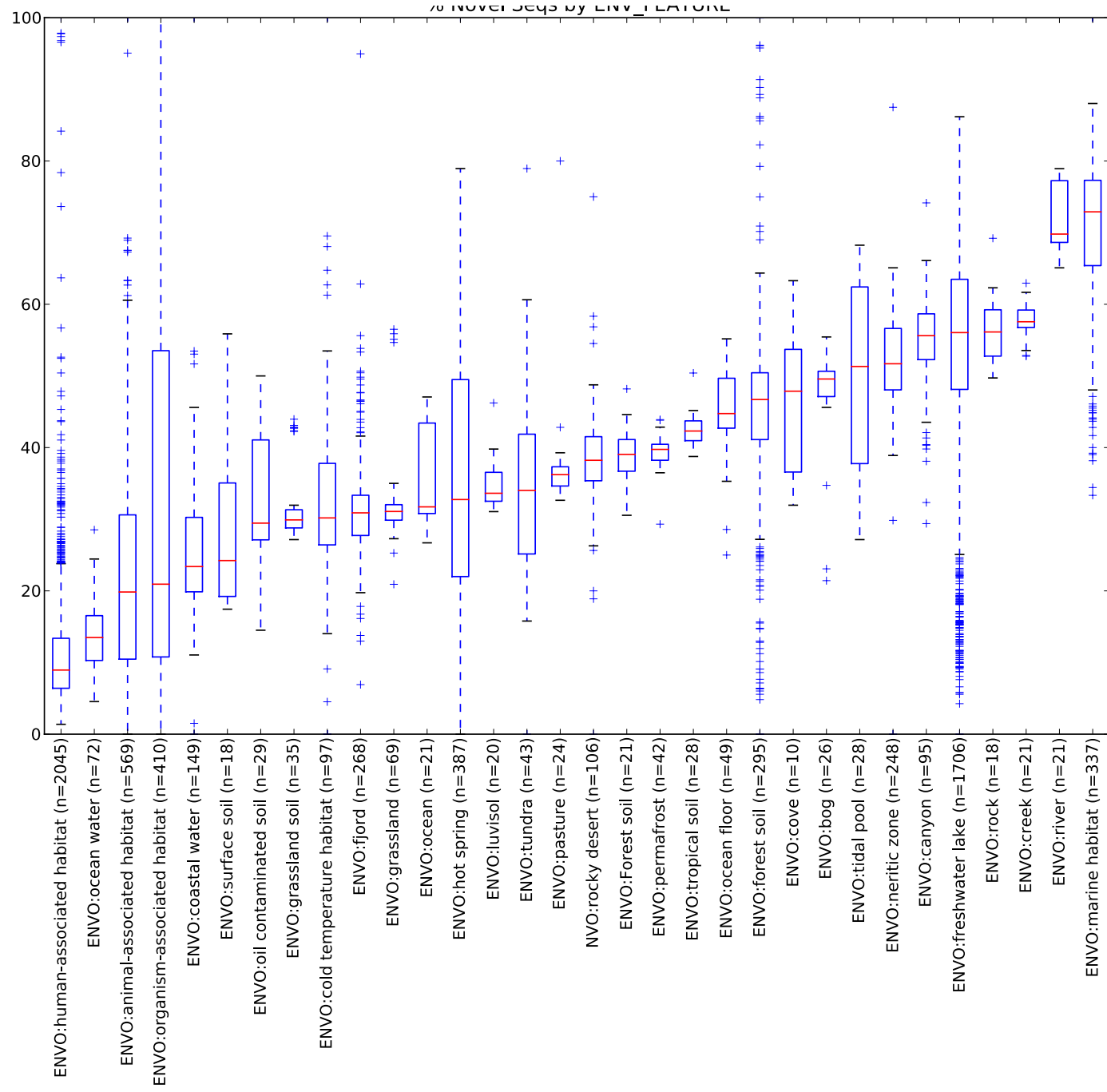
- Built-in quality filter
- Easily parallelizable

Cons

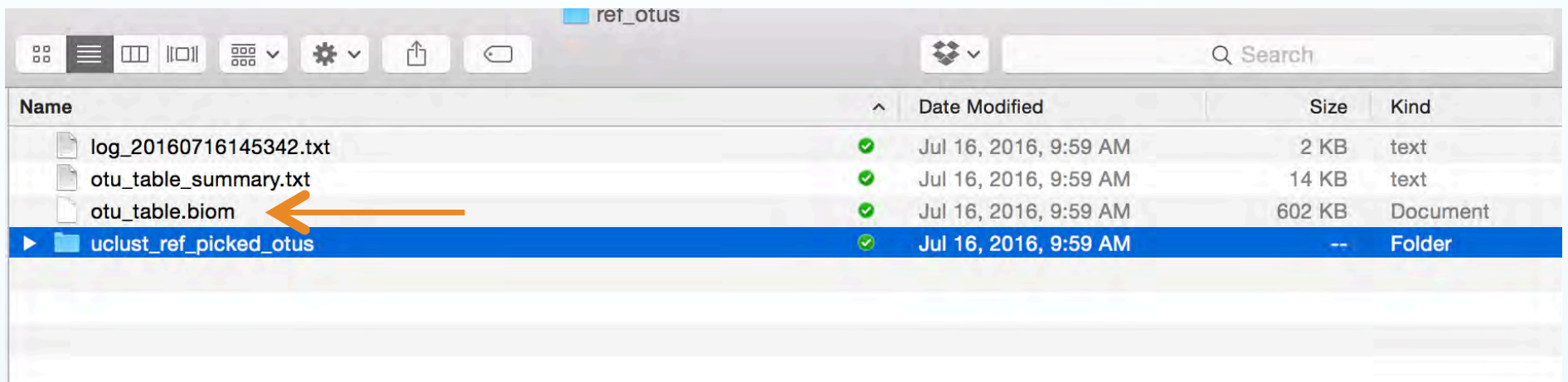
- Reference database bias
- Never observe new OTUs

```
$ pick_closed_reference_otus.py -i subs_seqs_q.fasta -r  
$PWD/gg_13_8_otus/rep_set/97_otus.fasta -t $PWD/  
gg_13_8_otus/taxonomy/97_otu_taxonomy.txt -o_ref_otus
```

Percentage of
reads that do not hit
the reference
collection, by
environment type.



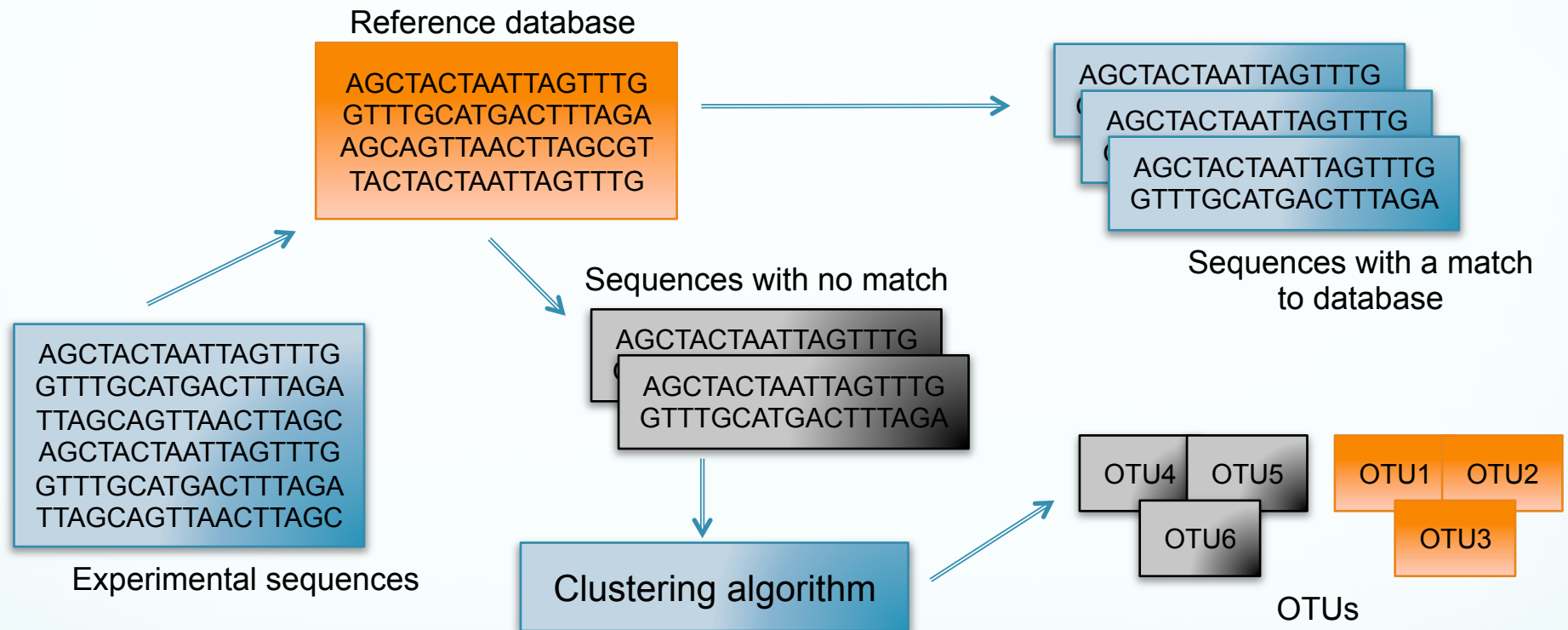
3.2 Closed-reference OTU picking



The screenshot shows a file explorer window titled 'ref_otus'. The window contains a list of files and folders with columns for Name, Date Modified, Size, and Kind. An orange arrow points to the file 'otu_table.biom'.

Name		Date Modified	Size	Kind
log_20160716145342.txt	✓	Jul 16, 2016, 9:59 AM	2 KB	text
otu_table_summary.txt	✓	Jul 16, 2016, 9:59 AM	14 KB	text
otu_table.biom	✓	Jul 16, 2016, 9:59 AM	602 KB	Document
▶ uclust_ref_picked_otus	✓	Jul 16, 2016, 9:59 AM	--	Folder

3.3 Open-reference OTU picking



Pros

- All reads are clustered
- Partially parallelizable

Cons

Mix of high quality sequences defining OTUs and possible low quality sequences defining OTUs

```
$ pick_open_reference_otus.py -i subs_seqs_q.fasta -r $PWD/  
gg_13_8_otus/rep_set/97_otus.fasta -o uclust_open_otu
```

3.3 Open-reference OTU picking

uclust_open_otu

Search

Name		Date Modified	Size	Kind
final_otu_map_mc2.txt	✓	Jul 16, 2016, 10:29 AM	107 KB	text
final_otu_map.txt	✓	Jul 16, 2016, 10:29 AM	216 KB	text
index.html	✓	Jul 16, 2016, 10:29 AM	2 KB	HTML text
log_20160716151935.txt	✓	Jul 16, 2016, 10:29 AM	4 KB	text
new_refseqs.fna	✓	Jul 16, 2016, 10:29 AM	143.3 MB	TextEd...ument
otu_table_mc2_summary.txt	✓	Jul 16, 2016, 10:28 AM	14 KB	text
otu_table_mc2_w_tax_no_pynast_failures.biom	✓	Jul 16, 2016, 10:28 AM	370 KB	Document
otu_table_mc2_w_tax.biom	✓	Jul 16, 2016, 10:28 AM	370 KB	Document
otu_table_mc2.biom	✓	Jul 16, 2016, 10:28 AM	132 KB	Document
▶ pynast_aligned_seqs	✓	Jul 16, 2016, 10:28 AM	--	Folder
rep_set.fna	✓	Jul 16, 2016, 10:28 AM	269 KB	TextEd...ument
rep_set.tre	✓	Jul 16, 2016, 10:28 AM	33 KB	Document
▶ step1_otus	✓	Jul 16, 2016, 10:27 AM	--	Folder
▶ step4_otus	✓	Jul 16, 2016, 10:27 AM	--	Folder
▶ uclust_assigned_taxonomy	✓	Jul 16, 2016, 10:27 AM	--	Folder

BIOM files



- Summarizing biom files

```
$ biom summarize-table -i uclust_otus/otu_table_even10.biom
```

```
$ biom summarize-table -i uclust_otus/otu_table_even10.biom --qualitative
```

- Adding metadata

```
$ biom add-metadata -i $PWD/otu_table_even10.biom -o  
denovo_otu.biom --sample-metadata-fp map_file.txt --observation-  
metadata-fp $PWD/uclust_assigned_taxonomy/  
subs_seqs_q_rep_set_tax_assignments.txt --observation-header  
OTUID,taxonomy --sc-separated taxonomy
```

- Convert to text file

```
$ biom convert -i otu_table.biom -o otu_table.txt --table-type "otu table" --  
header-key taxonomy -b
```

OTU table (.txt)

Sample OTU matrix

Sample identifiers



```
1 # Constructed from biom file
2 #OTU ID PC.354 PC.355 PC.356 PC.481 PC.593 PC.607 PC.634 PC.635 PC.636 taxonomy
3 denovo0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 k_Bacteria; p_Firmicutes; c_Bacilli; o_Bacillales; f_Staphylococcaceae; g_Staphylococcus
4 denovo1 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae
5 denovo2 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 k_Bacteria; p_Bacteroidetes; c_Bacteroidia; o_Bacteroidales; f_S24-7; g__; s__
6 denovo3 0.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Bacteroidetes; c_Bacteroidia; o_Bacteroidales; f_S24-7; g__; s__
7 denovo4 0.0 0.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
8 denovo5 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 1.0 k_Bacteria; p_Bacteroidetes; c_Bacteroidia; o_Bacteroidales; f_S24-7; g__; s__
9 denovo6 1.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
10 denovo7 0.0 0.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
11 denovo8 1.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 1.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
12 denovo9 0.0 0.0 0.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia
13 denovo10 0.0 0.0 0.0 1.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
14 denovo11 0.0 0.0 0.0 1.0 0.0 0.0 0.0 0.0 1.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
15 denovo12 0.0 1.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 k_Bacteria; p_Firmicutes; c_Clostridia; o_Clostridiales; f_Lachnospiraceae; g__; s__
```



OTU identifiers



OTU information

www.QIIME.org

1

Sequencing output
(454, Illumina, Sanger)

fastq, fasta, qual, or sff/trace files

Metadata

mapping file

Pre-processing

e.g., remove primer(s), demultiplex,
quality filter

Denoise 454 Data

PyroNoise, Denoiser

Database Submission

(In development)

2

Pick OTUs and representative sequences

Reference based
BLAST, UCLUST,
USEARCH

De novo
e.g., UCLUST, CD-HIT,
MOTHUR, USEARCH

Assign taxonomy

BLAST, RDP
Classifier

Align sequences

e.g., PyNAST,
INFERNAL, MUSCLE,
MAFFT

Build 'OTU table'

i.e., sample by observation
matrix

Build phylogenetic tree

e.g., FastTree, RAXML,
ClearCut

3

OTU (or other sample by
observation) table

Phylogenetic Tree

Evolutionary relationship
between OTUs

α -diversity and rarefaction

e.g., Phylogenetic
Diversity, Chao1,
Observed Species

β -diversity and rarefaction

e.g., Weighted and
unweighted UniFrac, Bray-
Curtis, Jaccard

Interactive visualizations

e.g., PCoA plots, distance histograms, taxonomy charts, rarefaction
plots, network visualization, jackknifed hierarchical clustering.

Legend

Currently supported for
marker-gene data only

(i.e., 'upstream' step)

Required step or input

Currently supported for
general sample by
observation data

(i.e., 'downstream' step)

Optional step or input

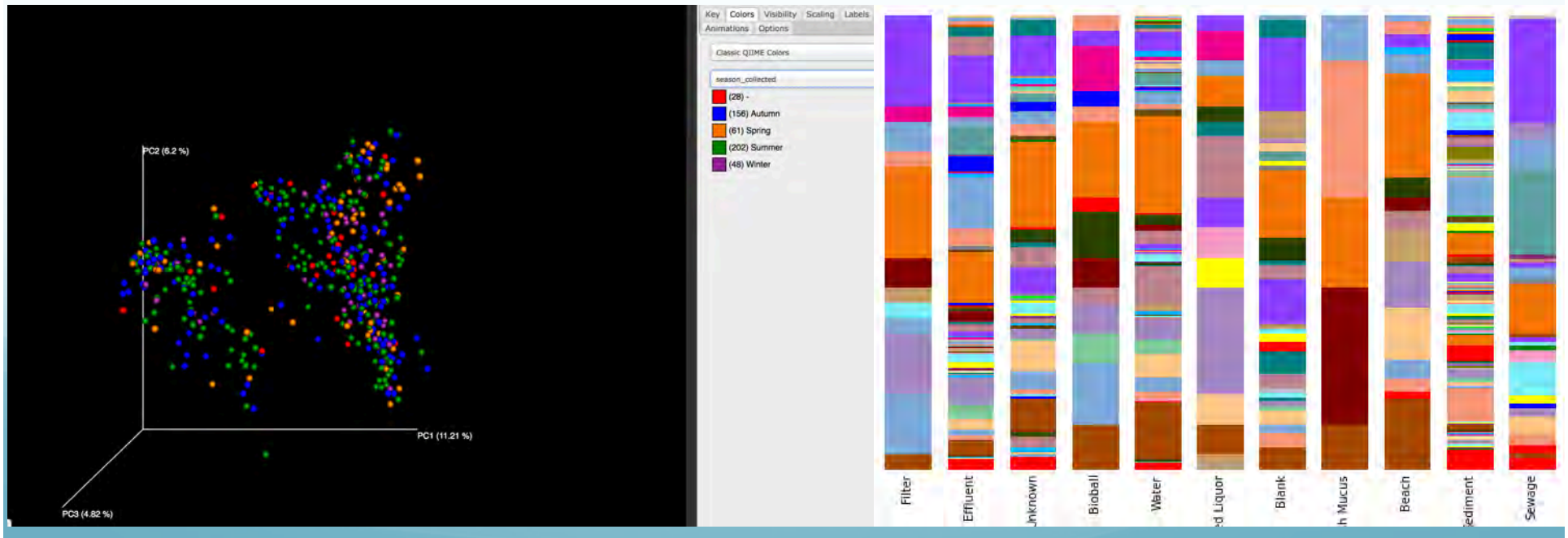
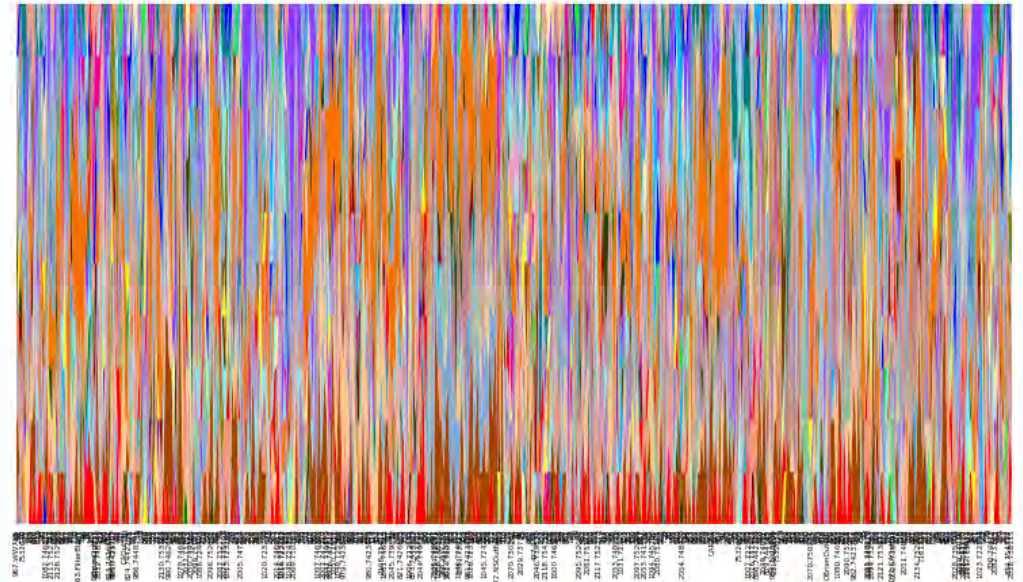
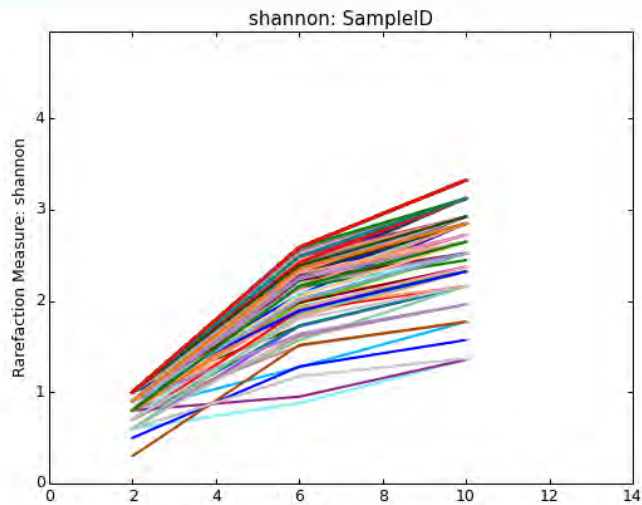
Watch out for this!

A note on chimera removal

- QIIME uses ChimeraSlayer to detect chimeric sequences using your alignment and a reference database
- Remove these OTUs from your OTU table and alignment before proceeding with tree building and visualization of results

```
$ identify_chimeric_seqs.py -m ChimeraSlayer -i rep_set_aligned.fasta -a  
gold.fa -o chimeric_seqs.txt
```

```
$ core_diversity_analyses.py -i denovo_otu.biom -o core_diversity -e 10 -m map_file.txt -t ref_set.tre
```



1. Filter low sequence count samples from table (minimum sequence count: 7500)

```
$ filter_samples_from_otu_table.py -i denovo_otu.biom -o core_diversity7500/table_mc7500.biom -n 7500
```

2. Rarefy the OTU table to 7500 sequences/sample

```
$ single_rarefaction.py -i core_diversity7500/table_mc7500.biom -o core_diversity7500s/table_even7500.biom -d 7500
```

3. Beta Diversity (weighted_unifrac)

```
$ beta_diversity.py -i core_diversity7500s/table_even7500.biom -o core_diversity7500/bdiv_even7500/ --metrics weighted_unifrac -t ref_set.tre
```

4. Principal coordinates (weighted_unifrac)

```
$ principal_coordinates.py -i core_diversity7500/bdiv_even7500//weighted_unifrac_dm.txt -o core_diversity7500/bdiv_even7500//weighted_unifrac_pc.txt
```

5. Make emperor plots, weighted_unifrac)

```
$ make_emperor.py -i core_diversity7500s/bdiv_even7500//weighted_unifrac_pc.txt -o core_diversity7500/bdiv_even7500//weighted_unifrac_emperor_pcoa_plot/ -m map_file.txt
```

6. Alpha rarefaction

```
$ multiple_rarefactions.py -i core_diversity7500/table_mc7500.biom -m 10 -x 7500 -s 749 -o core_diversity7500/arare_max7500//rarefaction/
```

7. Alpha diversity on rarefied OTU tables

```
$ alpha_diversity.py -i core_diversity7500/arare_max7500//rarefaction/ -o core_diversity7500_1105noblanks/arare_max7500//alpha_div/ -t ref_set.tre
```

8. Collate alpha

```
$ collate_alpha.py -i core_diversity7500/arare_max7500//alpha_div/ -o core_diversity7500_/arare_max7500//alpha_div_collated/
```

9. Rarefaction plot: All metrics

```
$ make_rarefaction_plots.py -i core_diversity7500/arare_max7500//alpha_div_collated/ -m 150701_CAWSMF_1104.txt -o core_diversity7500/arare_max7500//alpha_rarefaction_plots/
```

10. Summarize Taxonomy

```
$ summarize_taxa.py -i core_diversity7500s/taxa_plots/table_mc7500_sorted.biom -o core_diversity7500/taxa_plots/
```

Resources and help

- QIIME offers a suite of developer-designed [tutorials](#).
- Documentation for all [QIIME scripts](#) is available.
- There is a very active [QIIME Forum](#) on Google.
- The [QIIME Blog](#) provides updates on bug fixes, new features and new releases (QIIME 2.0)